TEAM ALPHA — HACKATHON SUBMISSION

AccessLearn

An Inclusive, Multi-Modal Learning Platform for Everyone

Understanding the Challenge

Traditional education and e-learning platforms are built with a "one-size-fits-all" approach. They overwhelmingly rely on walls of text and long video lectures. For millions of students—including those with dyslexia, low vision, ADHD, or differing language proficiencies—this creates an invisible barrier to learning. When study material is difficult to digest, students lose engagement, fall behind, and experience learning fatigue.

> **Core Vision:** To ensure that every single student, regardless of their reading ability or learning disability, can understand complex academic concepts on their own terms.

Who is Affected?

Students with Reading Difficulties

Struggle with dense paragraphs and dyslexia, needing simplified vocabulary and clean, dyslexia-friendly typography.

Visual & Auditory Learners

Benefit immensely from interactive diagrams, color-coded mindmaps, and text-to-speech audio read-alouds.

Teachers & Educators

Lack easy tools to adapt standard curriculum into multiple accessible formats for diverse classrooms.

Parents & Guardians

Need clear visibility into their child's daily learning progress, study streaks, and areas requiring extra support.

Proposed Solution & User Workflow

Our Solution: "One Concept, Many Ways"

AccessLearn solves the accessibility crisis by transforming any lesson into five distinct, interchangeable learning formats. Instead of forcing a student to read difficult text, they can switch views instantly:

- **Simple Text View:** Strips away academic jargon into clear, elementary explanations with bullet points.
- **Visual Diagram View:** Interactive visual cards and structural breakdowns for visual thinkers.
- **Audio Read-Aloud:** Built-in text-to-speech engine that reads lessons aloud naturally.
- **Real-World Examples:** Connects abstract theories to everyday analogies and stories.
- **Practice Quiz:** Instant formative assessment to test comprehension immediately.

Complete Product Workflow

The user journey is designed to be frictionless, engaging, and entirely cloud-synchronized:

1. **User Authentication:** Students or teachers securely sign up or log in via Supabase Auth, maintaining secure session persistence across devices.
2. **Onboarding & Preferences:** Users set their preferred language (English or Hindi), text scaling, and accessibility preferences.
3. **Lesson Hub & Navigation:** Students pick a lesson (e.g., Photosynthesis, Recursion, Fractions) and explore the 5 multi-modal tabs.
4. **Intelligent Tutoring:** If a student has questions, they can chat with the floating **AI Study Buddy** chatbot, powered securely by backend serverless functions and Gemini AI.
5. **Progress Tracking:** Every quiz answered correctly rewards XP points, updates mastery percentages, and extends daily study streaks in real time.

Core Features & Technology Architecture

Main Features Built

- **Adaptive Multi-Modal Engine:** Instant switching between text, visuals, audio, examples, and quizzes.
- **AI Study Buddy Chatbot:** A floating interactive assistant that breaks down complex questions into simple terms.
- **Teacher Content Portal:** Empowers educators to publish custom lessons and quizzes that instantly sync for all students.
- **PDF Study Material Parser:** Students can drop study PDFs into the app to extract text and analyze content.
- **Parent & Teacher Bridge:** Analytics dashboards powered by Recharts showing student growth and quiz performance.

Technology Stack

Layer	Technology	Purpose in AccessLearn
Frontend	React 18, Vite, Tailwind CSS	Delivers a lightning-fast, responsive user interface with dynamic themes.
Backend & Auth	Supabase BaaS, PostgreSQL	Manages user login, encrypted sessions, and relational database tables.
Security	Row Level Security (RLS)	Ensures strict data privacy so users can only access their own private progress.
Serverless AI	Supabase Edge Functions, Gemini	Proxies AI chat requests securely without exposing API keys in the browser.

Execution Plan, Impact & Team Details

Development Plan Across Rounds

Hackathon Phase	Key Milestones & Deliverables
Round 1: Foundation	Built core React frontend, Atkinson typography, routing, and Supabase Auth integration.
Round 2: Core Functionality	Implemented database synchronization, Teacher Portal CRUD operations, and multi-modal tabs.
Round 3: Polish & AI	Deployed Supabase Edge Functions for the AI Study Buddy, refined UI contrast, and tested production build.

Expected Impact & Future Scope

Impact: AccessLearn democratizes education by removing cognitive bottlenecks. Students grasp concepts 3x faster through personalized formats, boosting confidence and academic retention.

Future Scope: Post-event, we plan to introduce live multi-user virtual study rooms, OCR for scanning physical textbook pages, and support for regional global languages.

Team & Participant Details

Team Name: Alpha

Project Title: AccessLearn (Inclusive E-Learning Platform)

Team Members & Responsibilities:

- **Akriti Anand (Team Leader):** Designed and developed the complete frontend architecture, user interface components, accessibility layouts, and multi-modal learning tabs.
- **Kartik (Team Member):** Built the backend cloud architecture, Supabase database configuration, Row Level Security (RLS) policies, and the AI integration using Supabase Edge Functions and Google Gemini.